



Green University of Bangladesh
Department of Computer Science and Engineering (CSE)
Faculty of Sciences and Engineering
Semester: (Spring, Year:2024), B.Sc. in CSE (Day)

Lab Report NO: 08

Course Title: Mobile Application Lab
Course Code: CSE-426 **Section: 211 D2**

Experiment Name: Implementation of Broadcast Intent and Broadcast Receiver

Student Details

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Submission Date: 23/05/2023

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Lab Report Status

Marks:

Signature:

Comments:

Date:

1. Title: Implementation of Broadcast Intent and Broadcast Receiver

2. OBJECTIVE:

- To Learning the concept of Broadcast Intents in the Android operating system.
- To Understanding different types of Broadcasts: system-generated and custom broadcasts.
- To Learning how to implement Broadcast Receivers to listen for and respond to broadcast intents.
- To Implementing code to register and unregister Broadcast Receivers dynamically at runtime.

3. PROBLEM STATEMENT:

❖ Concept of Broadcast Receivers:

- Broadcast Receivers are Android components that respond to broadcast messages (intents) from other applications or the system.
- They remain inactive until triggered by an incoming broadcast intent.

❖ Mechanism of Broadcast Intents:

- Broadcast intents are used to send messages to multiple components.
- These intents can be either system-generated (e.g., battery low) or custom-defined by applications.

❖ Detection and Registration:

- Broadcast Receivers detect broadcasts by being registered to listen for specific action strings in intents.
- Registration can be static (in the AndroidManifest.xml) or dynamic.

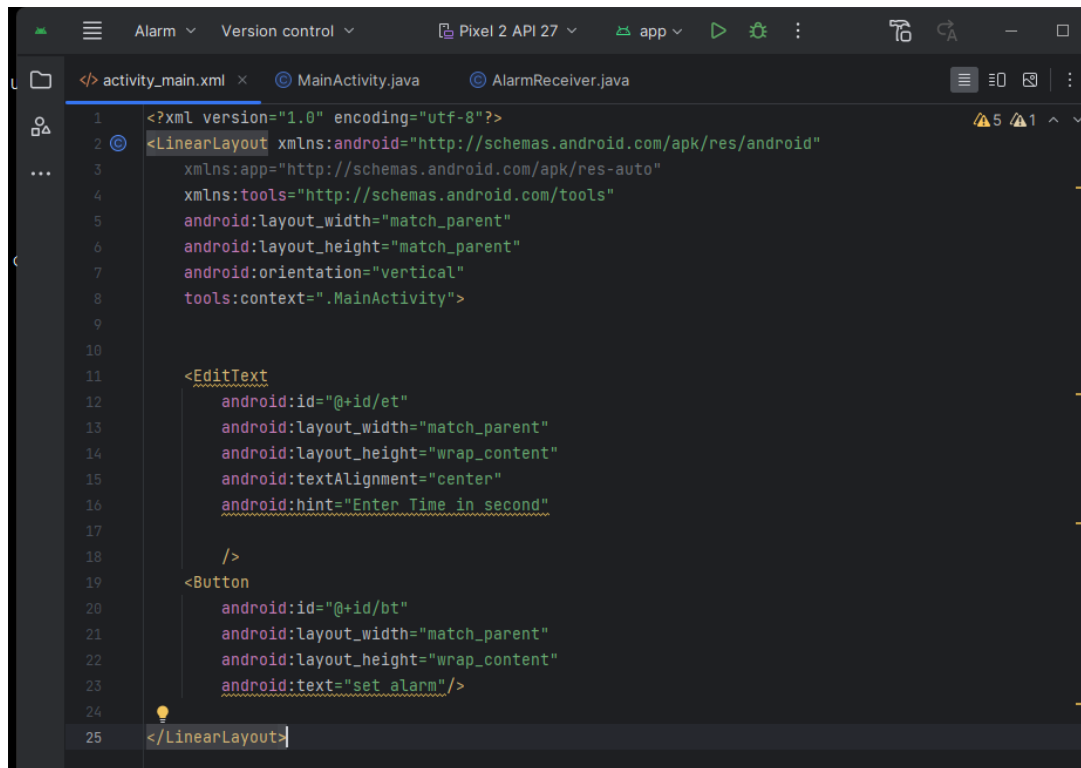
❖ Dormancy and Activation:

- Broadcast Receivers are dormant and consume minimal resources until activated by a matching intent.
- When a matching intent is broadcasted, the system wakes up the relevant Broadcast Receiver to handle the intent.

❖ Experiment Objective:

- The experiment involves creating an alarm clock application.
- The alarm clock will send a broadcast intent at a specified time.

4. Implementation of Broadcast for making an alarm clock:(XML File of Alarm Clock)



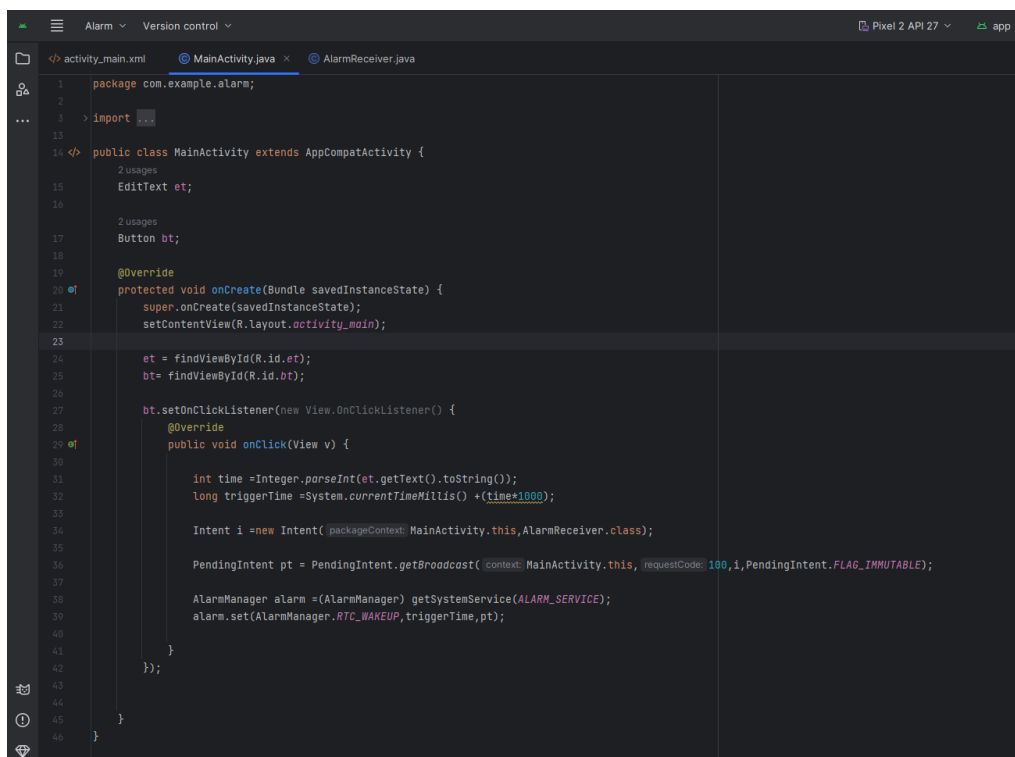
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/et"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textAlignment="center"
        android:hint="Enter Time in second"
    />

    <Button
        android:id="@+id/bt"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="set alarm"/>

</LinearLayout>
```

Main Activity Java File implementing an Alarm Clock:



```
package com.example.alarm;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.widget.EditText;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    EditText et;
    Button bt;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        et = findViewById(R.id.et);
        bt = findViewById(R.id.bt);

        bt.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                int time = Integer.parseInt(et.getText().toString());
                long triggerTime = System.currentTimeMillis() + (time*1000);

                Intent i = new Intent( packageContext: MainActivity.this, AlarmReceiver.class);

                PendingIntent pt = PendingIntent.getBroadcast( context: MainActivity.this, requestCode: 100, i, PendingIntent.FLAG_IMMUTABLE);

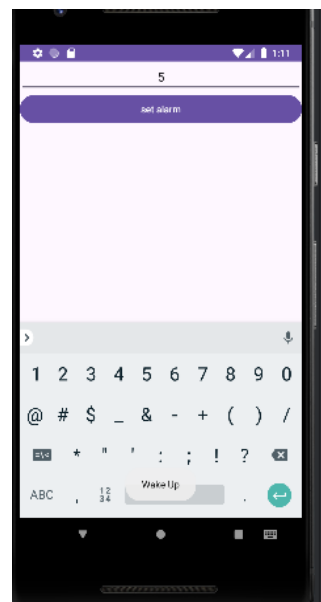
                AlarmManager alarm = (AlarmManager) getSystemService(ALARM_SERVICE);
                alarm.set(AlarmManager.RTC_WAKEUP, triggerTime, pt);

            }
        });
    }
}
```

Alarm Java File implementing Broadcast Receiver for an Alarm Clock:

```
Alarm Version control
activity_main.xml MainActivity.java AlarmReceiver.java x
1 package com.example.alarm;
2
3 > import ...
4
5
6
7
8
9
10 </> public class AlarmReceiver extends BroadcastReceiver {
11
12     @Override
13     public void onReceive(Context context, Intent intent) {
14
15         MediaPlayer mp = MediaPlayer.create(context, Settings.System.DEFAULT_RINGTONE_URI);
16         mp.setLooping(true);
17         mp.start();
18         Toast.makeText(context, text: "Wake Up", Toast.LENGTH_SHORT).show();
19
20
21
22     }
23 }
```

5. OUTPUT:



6. **DISCUSSION:** In this lab exercise, we have learned how to use a broadcast intent and broadcast receiver. And also learn how to implement broadcast receivers to listen for and respond to broadcast intents.